Undergraduate Thesis Proposal

Yunfan Du
School of Artificial Intelligence and Automation
Supervisor: Zhihuan Chen

I. Basic Information

Student Name	Yunfan Du
Student ID	202204416388
Affiliation	School of Artificial Intelligence and Automation, Robotics Engineering
Title	Cross-Terrain Quadruped Locomotion via Proprioceptive Adaptive Reward Scheduling
Supervisor	Zhihuan Chen
Attachment	None

II. Research Motivation and Significance

Quadruped robots provide superior mobility over wheeled and tracked platforms in unstructured environments, which makes them highly valuable for search and rescue, field inspection, and special operations. A central challenge is how to achieve an optimal trade-off between locomotion efficiency and stability across terrains with different difficulty levels.

Existing Hybrid Internal Model (HIM) learning frameworks usually rely on globally fixed weights in multi-objective reward functions. Although this design is simple and stable in training, it causes clear performance compromises: conservative behavior and reduced speed/energy efficiency on flat terrain, and risky actions with increased fall rate on highly rugged terrain due to persistent velocity incentives. More importantly, fixed priorities cannot adapt online to terrain changes, limiting the achievable cross-terrain performance.

To address this limitation, this proposal introduces a proprioception-driven adaptive reward-weight scheduler. Without modifying the core policy architecture or the two-stage training paradigm, the method estimates terrain difficulty online from proprioceptive signals and dynamically reallocates reward-term weights. The expected outcome is improved efficiency on traversable terrain and enhanced robustness on high-risk terrain, leading to better overall locomotion performance.

III. Related Work and Trends

1. International Research

Representative directions include:

- learning-based gait generation, velocity tracking, and disturbance rejection;
- domain randomization and system identification for better sim-to-real transfer;
- integration of perception and control for terrain-adaptive locomotion;
- curriculum learning and adaptive objective design for stronger generalization.

2. Domestic Research

Recent domestic studies have advanced rapidly in legged robot hardware, motion planning, and reinforcement-learning-based control. Current focuses include gait/energy optimization, hybrid control with model-based priors, and engineering-oriented robustness and real-time deployment.

3. Development Trends

Future trends include:

1. more efficient and stable training paradigms;
2. tighter integration of perception, decision-making, and control;
3. improved sim-to-real reliability with lower deployment cost;
4. safety-aware and interpretable locomotion policies.

IV. Research Content and Methodology

1. Research Content

    1. Reproduce and analyze the HIM learning framework to establish fixed-weight performance baselines.
    2. Design terrain-difficulty indicators from proprioceptive observations.
    3. Implement and integrate an adaptive reward scheduler without changing core network architecture and training pipeline.
    4. Evaluate speed, energy cost, stability, and fall rate under multiple terrains and disturbances.
    5. Conduct comparative and ablation studies to validate dynamic efficiency-stability balancing.

2. Methodology

    - Literature review;
    - simulation-based experimentation;
    - quantitative comparative analysis;
    - ablation analysis of scheduler design and key hyperparameters.

V. Timeline

Period	Plan
Week 1	Literature review, problem definition, and proposal drafting.
Week 2	Baseline reproduction and simulation environment setup.
Weeks 3–4	Design, implementation, and integration of terrain estimator and adaptive scheduler.
Week 5	Comparative and ablation experiments on multiple terrains.
Week 6	Thesis writing for method, experiments, and analysis.
Week 7	Final revision, similarity check, and defense preparation.

VI. Expected Outcomes

1. A proprioception-driven adaptive reward-weight scheduling method integrated with the HIM framework.
2. Complete comparative and ablation results between fixed-weight and adaptive-weight settings.